

Module 2

Homogeneous Equations, Linear combinations, Linear Independence/Dependence, Span, Basis, Dimension, Rank, Subspaces, Column Space, Rank-Nullity Theorem.

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Chapter 1

Homogeneous Equations

1.0.1 General Theory

A system of linear equations is called **homogeneous** if it is of the form

$$AX = 0,$$

where A is the coefficient matrix, X is the vector of unknowns, and 0 is the zero vector.

For a 3×3 homogeneous system,

$$\begin{aligned}a_1x + b_1y + c_1z &= 0 \\a_2x + b_2y + c_2z &= 0 \\a_3x + b_3y + c_3z &= 0,\end{aligned}$$

the coefficient matrix is

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}.$$

Step 1: Determinant Test

Compute $\det(A)$.

- If $\det(A) \neq 0$, then the system has only the trivial solution:

$$x = y = z = 0.$$

- If $\det(A) = 0$, then the system has infinitely many solutions.

Remark

A homogeneous system is always consistent because

$$x = y = z = 0$$

is always a solution.

Hence, an inconsistent homogeneous system does not exist.

Step 2: Gaussian Elimination

After determining $\det(A)$, reduce the augmented matrix

$$[A|0]$$

to row echelon form using Gaussian elimination, and obtain the solution set.

1.0.2 Problems on Homogeneous system of equations

Problem 1

$$\begin{aligned}x + y + z &= 0 \\2x + 3y + 4z &= 0 \\3x + 5y + 6z &= 0\end{aligned}$$

Problem 2

$$\begin{aligned}x + y + z &= 0 \\2x + 2y + 2z &= 0 \\3x + 3y + 3z &= 0\end{aligned}$$

Solution to Problem 2

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 \\ 3 & 3 & 3 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Here, rank is $r = 1$ and the number variables is $n = 3$.

So, there are $n - r = 3 - 1 = 2$ free variables.

Therefore, let $y = s$, $z = t$.

Then, from $x + y + z = 0$, we have

$$x = -s - t$$

$$(x, y, z) = (-s - t, s, t)$$

Infinitely many solutions.

Problem 3

$$\begin{aligned}2x + y - z &= 0 \\4x + 2y - 2z &= 0 \\x + y + z &= 0\end{aligned}$$

Solution to Problem 3

Step 1: Write the Augmented Matrix

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 4 & 2 & -2 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right]$$

Step 2: Convert to Row Echelon Form

Interchange R_1 and R_3 :

$$\begin{aligned}R_1 &\leftrightarrow R_3 \\ \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 4 & 2 & -2 & 0 \\ 2 & 1 & -1 & 0 \end{array} \right] \end{aligned}$$

Eliminate entries below first pivot:

$$\begin{aligned}R_2 &\rightarrow R_2 - 4R_1 \\ R_3 &\rightarrow R_3 - 2R_1 \\ \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & -6 & 0 \\ 0 & -1 & -3 & 0 \end{array} \right] \end{aligned}$$

Make second pivot equal to 1:

$$R_2 \rightarrow -\frac{1}{2}R_2$$
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & -1 & -3 & 0 \end{array} \right]$$

Eliminate entry in R_3 :

$$R_3 \rightarrow R_3 + R_2$$
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Step 3: Interpretation

The last row is:

$$0 = 0$$

This shows that one equation is dependent on the others.

Hence, the system has infinitely many solutions.

Step 4: Identify Free Variable

From the matrix, we get:

$$x + y + z = 0$$
$$y + 3z = 0$$

There are three variables and only two independent equations. So, number of free variables:

$$3 - 2 = 1$$

Let

$$z = t \quad (t \in \mathbb{R})$$

Step 5: Find General Solution

From $y + 3z = 0$:

$$y = -3z = -3t$$

From $x + y + z = 0$:

$$x + (-3t) + t = 0$$

$$x = 2t$$

Step 6: General Solution

$$\boxed{x = 2t, \quad y = -3t, \quad z = t}$$

where $t \in \mathbb{R}$.

Problem 4

$$2x + y + z = 0$$

$$x + 2y + z = 0$$

$$x + y + 2z = 0$$

Problem 5

$$x + 2y + 3z = 0$$

$$2x + 4y + 6z = 0$$

$$3x + 6y + 9z = 0$$

Chapter 2

Linear Combination Theory

2.1 Introduction

Linear algebra is one of the fundamental pillars of modern engineering mathematics. Almost every area of science and engineering makes extensive use of vectors and vector spaces.

The concept of *linear combination* lies at the heart of vector space theory. It describes how vectors can be constructed from simpler vectors using scalar multiplication and vector addition.

In engineering applications, linear combinations appear in:

- Signal processing and superposition of waves,
- Structural analysis and force resolution,
- Electrical circuits and current distribution,
- Machine learning and feature representation,
- Quantum mechanics and state superposition.

Understanding linear combinations is therefore essential for advanced study in mathematics, physics, and engineering.

2.2 Vector Spaces: A Brief Review

Before studying linear combinations, it is necessary to recall the concept of a vector space.

Definition 2.1. A non-empty set V together with two operations called vector addition and scalar multiplication is called a **vector space** over a field F if the following axioms are satisfied:

1. Closure under addition,
2. Commutativity of addition,
3. Associativity of addition,
4. Existence of zero vector,
5. Existence of additive inverse,
6. Closure under scalar multiplication,
7. Distributive laws,
8. Existence of scalar identity.

Typical examples of vector spaces include:

- $\mathbb{R}^n$ (real coordinate space),
- Polynomial spaces,
- Function spaces,
- Matrix spaces.

All discussions in this chapter take place within a vector space.

2.3 Definition of Linear Combination

Definition 2.2. Let V be a vector space over a field F , and let

$$v_1, v_2, \dots, v_n \in V.$$

A vector $v \in V$ is said to be a **linear combination** of $v_1, v_2, \dots, v_n$ if there exist scalars $a_1, a_2, \dots, a_n \in F$ such that

$$v = a_1v_1 + a_2v_2 + \dots + a_nv_n.$$

The scalars $a_1, a_2, \dots, a_n$ are called the **coefficients** of the linear combination.

2.4 Theoretical Interpretation

A linear combination represents the process of constructing new vectors from existing ones.

The two fundamental operations involved are:

1. Scaling: multiplying a vector by a scalar,
2. Addition: adding scaled vectors.

Thus, linear combinations describe all vectors that can be formed by stretching, shrinking, reversing, and adding given vectors.

If a vector can be expressed as a linear combination of given vectors, then it contains no new directional information beyond those vectors.

2.5 Geometric Interpretation

2.5.1 In $\mathbb{R}^2$

Consider two vectors $u, v \in \mathbb{R}^2$.

- If u and v are collinear, their linear combinations form a line.
- If u and v are not collinear, their linear combinations form a plane.

Thus,

$$au + bv \quad (a, b \in \mathbb{R})$$

represents all points in a line or plane.

2.5.2 In $\mathbb{R}^3$

For three vectors $u, v, w \in \mathbb{R}^3$:

- One vector generates a line,
- Two independent vectors generate a plane,
- Three independent vectors generate space.

Hence, linear combinations describe geometric objects.

2.6 Linear Combinations and Systems of Equations

Let

$$v = a_1v_1 + \cdots + a_nv_n.$$

This equation is equivalent to a system of linear equations in the unknowns $a_1, \dots, a_n$. Therefore, determining whether v is a linear combination of given vectors is equivalent to solving a system of linear equations.

2.6.1 Matrix Representation

Let

$$A = [v_1 \ v_2 \ \cdots \ v_n]$$

be a matrix whose columns are the given vectors.

Then,

$$Ax = v$$

where $x = (a_1, \dots, a_n)^T$.

Thus, linear combinations are directly related to matrix theory.

2.7 Example Problems

2.7.1 Example 1

Express the vector $(3, 5, 2)$ as a linear combination of the vectors $(1, 1, 0)$, $(2, 3, 0)$ and $(0, 0, 1)$ of $V_3(\mathbb{R})$.

Solution . Let

$$v_1 = (1, 1, 0), \quad v_2 = (2, 3, 0), \quad v_3 = (0, 0, 1)$$

Write the generic linear combination,

$$(3, 5, 2) = c_1 v_1 + c_2 v_2 + c_3 v_3$$

$$(3, 5, 2) = c_1(1, 1, 0) + c_2(2, 3, 0) + c_3(0, 0, 1)$$

$$(3, 5, 2) = (c_1 + 2c_2, c_1 + 3c_2, c_3)$$

Equating components gives

$$c_1 + 2c_2 = 3$$

$$c_1 + 3c_2 = 5$$

$$c_3 = 2$$

Solving,

$$c_2 = 2, \quad c_1 = -1, \quad c_3 = 2$$

Therefore

$$(3, 5, 2) = -1(1, 1, 0) + 2(2, 3, 0) + 2(0, 0, 1)$$

2.7.2 Example 2

Express the vector $(2, -1, -8)$ as a linear combination of the vectors $(1, 2, 1)$, $(1, 1, -1)$ and $(4, 5, -2)$ of $V_3(\mathbb{R})$.

Solution.

Let

$$v_1 = (1, 2, 1), \quad v_2 = (1, 1, -1), \quad v_3 = (4, 5, -2)$$

Assume

$$(2, -1, -8) = c_1 v_1 + c_2 v_2 + c_3 v_3$$

$$(2, -1, -8) = c_1(1, 2, 1) + c_2(1, 1, -1) + c_3(4, 5, -2)$$

$$(2, -1, -8) = (c_1 + c_2 + 4c_3, 2c_1 + c_2 + 5c_3, c_1 - c_2 - 2c_3)$$

Equating components

$$c_1 + c_2 + 4c_3 = 2$$

$$2c_1 + c_2 + 5c_3 = -1$$

$$c_1 - c_2 - 2c_3 = -8$$

This system can be written as

$$\begin{bmatrix} 1 & 1 & 4 \\ 2 & 1 & 5 \\ 1 & -1 & -2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -8 \end{bmatrix}$$

Row reduction gives

$$\left[\begin{array}{ccc|c} 1 & 1 & 4 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Hence the system has infinitely many solutions.

Let

$$c_3 = k$$

$$c_2 = 5 - 3k$$

$$c_1 = -3 - k$$

Therefore

$$(2, -1, -8) = (-3 - k)(1, 2, 1) + (5 - 3k)(1, 1, -1) + k(4, 5, -2)$$

2.7.3 Example 3

Verify whether the vector $(2, -5, 3)$ in $\mathbb{R}^3$ is a linear combination of the vectors $(1, -3, 2)$, $(2, -4, -1)$ and $(1, -5, 7)$.

Solution

Assume

$$(2, -5, 3) = c_1(1, -3, 2) + c_2(2, -4, -1) + c_3(1, -5, 7)$$

$$(2, -5, 3) = (c_1 + 2c_2 + c_3, -3c_1 - 4c_2 - 5c_3, 2c_1 - c_2 + 7c_3)$$

Equating components

$$c_1 + 2c_2 + c_3 = 2$$

$$-3c_1 - 4c_2 - 5c_3 = -5$$

$$2c_1 - c_2 + 7c_3 = 3$$

Matrix form

$$\begin{bmatrix} 1 & 2 & 1 \\ -3 & -4 & -5 \\ 2 & -1 & 7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix}$$

So, the augmented matrix is

$$[A|B] := \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ -3 & -4 & -5 & -5 \\ 2 & -1 & 7 & 3 \end{array} \right]$$

Using row operations (recall Gauss-elimination), the matrix $[A|B]$ reduces to

$$[A|B] \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 1 \\ 0 & 0 & 0 & 3 \end{array} \right]$$

Since

$$\text{rank}([A|B]) \neq \text{rank}(A)$$

the system is inconsistent.

Therefore $(2, -5, 3)$ is not a linear combination of the given vectors.

2.7.4 Example 4

Verify that

$$A = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$$

is a linear combination of

$$\begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

Solution

Assume

$$\begin{aligned}\begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix} &= c_1 \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} + c_2 \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} c_1 + c_2 + c_3 & -c_1 + c_2 + c_3 \\ -c_2 & -c_1 \end{bmatrix}\end{aligned}$$

Equating entries

$$c_1 + c_2 + c_3 = 3$$

$$-c_1 + c_2 + c_3 = -1$$

$$-c_2 = 1$$

$$-c_1 = -2$$

Thus

$$c_1 = 2, \quad c_2 = -1, \quad c_3 = 2$$

Therefore

$$\begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix} = 2 \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} + 2 \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

Hence the matrix is a linear combination of the given matrices

2.7.5 Practice problems

1. Express the vector $(4, 7, 3)$ as a linear combination of the vectors $(1, 1, 0)$, $(2, 1, 1)$ and $(0, 0, 1)$ in $V_3(\mathbb{R})$. That is, find scalars $c_1, c_2, c_3 \in \mathbb{R}$ such that

$$(4, 7, 3) = c_1(1, 1, 0) + c_2(2, 1, 1) + c_3(0, 0, 1).$$

2. Express the vector $(5, -2, 4)$ as a linear combination of the vectors $(1, 2, 0)$, $(2, 1, 1)$ and $(1, -1, 2)$ in $V_3(\mathbb{R})$. That is, determine c_1, c_2, c_3 such that

$$(5, -2, 4) = c_1(1, 2, 0) + c_2(2, 1, 1) + c_3(1, -1, 2).$$

3. Verify whether the vector $(3, -1, 5)$ in $\mathbb{R}^3$ is a linear combination of the vectors

$$(1, -2, 1), \quad (2, 1, 3), \quad (1, 0, 2).$$

If it is possible, express $(3, -1, 5)$ as a linear combination of these vectors.

4. Verify that the matrix

$$A = \begin{bmatrix} 4 & 2 \\ 1 & -3 \end{bmatrix}$$

is a linear combination of the matrices

$$B_1 := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B_2 := \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}, \quad B_3 := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

That is, find scalars c_1, c_2, c_3 such that

$$A = c_1 B_1 + c_2 B_2 + c_3 B_3.$$

2.8 Existence and Uniqueness

2.8.1 Existence

A vector v can be expressed as a linear combination of $v_1, \dots, v_n$ if and only if the system

$$Ax = v$$

is consistent.

2.8.2 Uniqueness

Theorem 2.1. *If $v_1, \dots, v_n$ are linearly independent, then every vector in their span has a unique representation as a linear combination¹.*

Proof. Suppose

$$v = a_1 v_1 + \dots + a_n v_n = b_1 v_1 + \dots + b_n v_n.$$

Then

$$(a_1 - b_1)v_1 + \dots + (a_n - b_n)v_n = 0.$$

Since the vectors are independent,

$$a_1 = b_1, \dots, a_n = b_n.$$

Hence the representation is unique. □

2.9 Applications in Engineering

Signal Decomposition: Any signal can be represented as a linear combination of basis signals.

Force Resolution: Forces are resolved into components using linear combinations.

Data Representation: Feature vectors in data science are expressed as linear combinations.

Quantum States: Quantum states are superpositions (linear combinations) of basis states.

¹This jumps the gun a bit. Span and linear independence are explained in the later chapters but it is good to know their relation with uniqueness of linear combination here.

2.10 Common Mistakes

- Ignoring consistency of equations,
- Making algebraic mistakes,
- Forgetting uniqueness conditions,
- Confusing span and independence.

Careful matrix reduction avoids these mistakes.

2.11 Summary

In this chapter, we studied:

- Definition of linear combinations,
- Geometric interpretation,
- Matrix formulation,
- Existence and uniqueness,
- Applications.

Note: Linear combinations form the foundation of all further topics in linear algebra.

Chapter 3

Linear Independence and Dependence

3.1 Introduction

The concept of linear independence is fundamental in understanding the structure of vector spaces. It determines whether a given collection of vectors contains redundant information.

In engineering applications, independence ensures that different signals, forces, or variables contribute uniquely to a system.

Dependence, on the other hand, indicates unnecessary repetition and can often be eliminated to simplify analysis.

This chapter develops theoretical and practical methods for identifying independent and dependent sets.

3.2 Definition of Linear Independence

Definition 3.1. Let V be a vector space over a field F . A set of vectors $\{v_1, v_2, \dots, v_n\} \subset V$ is said to be **linearly independent** if

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0$$

implies

$$a_1 = a_2 = \cdots = a_n = 0.$$

If there exist scalars, not all zero, satisfying the above equation, the vectors are said to be **linearly dependent**.

3.3 Conceptual Interpretation

Linear independence expresses the idea that no vector in the set can be written as a linear combination of the others.

3.3.1 Independent Sets

If vectors are independent:

- Each vector introduces a new direction,
- No vector is redundant,
- Information content is maximal.

3.3.2 Dependent Sets

If vectors are dependent:

- At least one vector can be removed,
- The span remains unchanged,
- The set contains redundancy.

3.4 Equivalent Characterizations

Theorem 3.1. *The following statements are equivalent for vectors $v_1, \dots, v_n$:*

1. *They are linearly independent.*
2. *The equation*

$$a_1v_1 + \dots + a_nv_n = 0$$

has only the trivial solution.

3. *No vector is a linear combination of the others.*
4. *Each vector has a unique representation in the span.*

Proof. The equivalence follows from basic properties of linear combinations and uniqueness of solutions of homogeneous systems. $\square$

3.5 Matrix Method for Testing Independence

Let

$$A = [v_1 \ v_2 \ \dots \ v_n]$$

be the matrix whose columns are the vectors.

Consider the homogeneous system

$$Ax = 0.$$

3.5.1 Criterion

Theorem 3.2. *The vectors $v_1, \dots, v_n$ are linearly independent if and only if*

$$Ax = 0$$

has only the trivial solution.

This occurs if and only if A has a pivot in every column.

3.6 Rank Criterion

Theorem 3.3. *Let $A = [v_1 \cdots v_n]$.*

Then $v_1, \dots, v_n$ are independent if and only if

$$\text{rank}(A) = n.$$

Proof. Independence implies nullity is zero. By Rank–Nullity Theorem,

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Hence rank equals n .

□

3.7 Geometric Interpretation

3.7.1 In $\mathbb{R}^2$

- Independent vectors are not parallel.
- Dependent vectors lie on same line.

3.7.2 In $\mathbb{R}^3$

- Two independent vectors form a plane.
- Three independent vectors form space.
- Dependent vectors lie in plane.

3.7.3 General Case

Independence corresponds to dimensional growth of span.

3.8 Fundamental Theorems

Theorem 3.4. *Any set containing the zero vector is linearly dependent.*

Proof. Let $v_1 = 0$. Then

$$1 \cdot v_1 + 0 \cdot v_2 + \cdots + 0 \cdot v_n = 0.$$

Non-trivial solution exists. $\square$

Theorem 3.5. *Any set of more than n vectors in $\mathbb{R}^n$ is dependent.*

Proof. Rank of any matrix in $\mathbb{R}^n$ is at most n . Hence more than n columns implies dependence. $\square$

Theorem 3.6. *Any subset of an independent set is independent.*

Proof. Removing vectors cannot create dependence. $\square$

3.9 Worked Examples

Example 3.1. Test independence of $(1, 2, 3)$, $(2, 4, 6)$, $(1, 1, 1)$.

Solution:

Form matrix:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 1 \end{bmatrix}$$

Row reduce:

$$\sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank = $2 < 3$.

Hence dependent.

Example 3.2. Check whether $(1, 0, 1)$, $(0, 1, 1)$, $(1, 1, 0)$ are independent.

Solution:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\det(A) = -2 \neq 0.$$

Hence independent.

Example 3.3. Determine independence of $(1, 1, 1, 1)$, $(2, 2, 2, 2)$, $(1, 0, 1, 0)$.

Solution: Second vector is multiple of first. Hence dependent.

Example 3.4. Check independence in polynomial space:

$$\{1 + x, x + x^2, 1 + x^2\}.$$

Solution:

Suppose

$$a(1 + x) + b(x + x^2) + c(1 + x^2) = 0.$$

Compare coefficients.

System has only trivial solution.

Hence independent.

Example 3.5. Test independence of functions:

$$\{e^x, e^{2x}, e^{3x}\}.$$

Solution:

Assume

$$ae^x + be^{2x} + ce^{3x} = 0.$$

Divide by e^x .

Polynomial in e^x .

Only trivial solution.

Hence independent.

3.10 Applications in Engineering

Signal Processing: Independent signals avoid interference.

Structural Mechanics: Independent forces determine stable structures.

Control Systems: Independent states improve controllability.

Data Science: Independent features reduce redundancy.

3.11 Common Errors

- Forgetting homogeneous condition,
- Incorrect row reduction,
- Confusing orthogonality with independence,
- Ignoring trivial solution.

3.12 Summary

This chapter established:

- Definition of independence,
- Matrix methods,
- Rank criterion,
- Geometric meaning,
- Engineering relevance.

Linear independence is essential for basis, dimension, and coordinate systems.

3.13 Practice Problems

Exercise 3.1. Test independence of $(2, 1, 0)$, $(1, 1, 1)$, $(0, 1, 2)$.

Exercise 3.2. Check whether $\{1, x, x^2\}$ is independent in P_2 .

Exercise 3.3. Determine whether $(1, 2, 3, 4)$, $(2, 4, 6, 8)$, $(1, 0, 0, 1)$ are independent.

Exercise 3.4. Test independence of $\{\sin x, \cos x, 1\}$.

Exercise 3.5. Find constants a, b, c such that

$$a(1, 2, 3) + b(2, 1, 0) + c(0, 1, 1) = 0.$$

Chapter 4

Span of a Set of Vectors

4.1 Introduction

The concept of span describes all vectors that can be generated from a given set of vectors using linear combinations.

It plays a central role in:

- Describing solution spaces of linear systems,
- Constructing vector spaces,
- Defining subspaces,
- Understanding dimension and basis.

In engineering mathematics, span determines whether a given set of vectors is sufficient to represent all possible states of a system.

This chapter develops the theory of span and its applications.

4.2 Definition of Span

Definition 4.1. Let V be a vector space over a field F and let $v_1, v_2, \dots, v_n \in V$.

The **span** of these vectors is defined as

$$\text{Span}\{v_1, v_2, \dots, v_n\} = \{a_1v_1 + a_2v_2 + \dots + a_nv_n \mid a_i \in F\}.$$

Thus, the span consists of all possible linear combinations of the given vectors.

4.3 Interpretation of Span

The span of a set of vectors represents all vectors that can be generated using those vectors.

4.3.1 Physical Interpretation

In physics and engineering:

- Forces span the space of possible motions,
- Signals span waveform spaces,
- Basis signals span communication channels.

4.3.2 Algebraic Interpretation

Algebraically, span is the smallest set containing the given vectors that is closed under addition and scalar multiplication.

4.4 Span as a Subspace

Theorem 4.1. *Let $S = \{v_1, \dots, v_n\} \subset V$. Then $\text{Span}(S)$ is a subspace of V .*

Proof. We verify the three subspace properties.

(i) **Zero vector:**

$$0 = 0v_1 + \dots + 0v_n \in \text{Span}(S).$$

(ii) **Closure under addition:**

Let

$$u = \sum a_i v_i, \quad w = \sum b_i v_i.$$

Then

$$u + w = \sum (a_i + b_i) v_i \in \text{Span}(S).$$

(iii) **Closure under scalar multiplication:**

For scalar c ,

$$cu = \sum (ca_i) v_i \in \text{Span}(S).$$

Hence, $\text{Span}(S)$ is a subspace. □

4.5 Smallest Subspace Property

Theorem 4.2. *The span of a set S is the smallest subspace of V containing S .*

Proof. Let W be any subspace containing S .

Since W is closed under linear combinations, it must contain all linear combinations of elements of S .

Therefore,

$$\text{Span}(S) \subseteq W.$$

Hence, span is the smallest such subspace. $\square$

4.6 Generating Sets

Definition 4.2. A set $S \subset V$ is called a **generating set** for V if

$$\text{Span}(S) = V.$$

A generating set produces the entire vector space.
Such sets need not be minimal.

4.7 Minimal Spanning Sets

A spanning set is said to be **minimal** if no vector can be removed without destroying the spanning property.

Theorem 4.3. *A spanning set is minimal if and only if it is linearly independent.*

Proof. If dependent, some vector is redundant.

If independent, removal destroys span.

Hence equivalence holds. $\square$

This theorem establishes a deep connection between span and independence.

4.8 Matrix Interpretation of Span

Let

$$A = [v_1 \ v_2 \ \cdots \ v_n].$$

Then

$$\text{Span}\{v_1, \dots, v_n\} = \text{Col}(A),$$

where $\text{Col}(A)$ denotes the column space of A .

Thus, span theory is closely related to matrix theory.

4.9 Geometric Interpretation

4.9.1 In $\mathbb{R}^2$

- One nonzero vector spans a line,
- Two independent vectors span the plane.

4.9.2 In $\mathbb{R}^3$

- One vector spans a line,
- Two independent vectors span a plane,
- Three independent vectors span space.

4.9.3 Higher Dimensions

Span determines dimensional structure of spaces.

4.10 Worked Examples

Example 4.1. Find the span of $(1, 0)$ and $(0, 1)$ in $\mathbb{R}^2$.

Solution:

$$a(1, 0) + b(0, 1) = (a, b).$$

All vectors of $\mathbb{R}^2$ are obtained.

Hence,

$$\text{Span} = \mathbb{R}^2.$$

Example 4.2. Find the span of $(2, 4)$.

Solution:

$$a(2, 4) = (2a, 4a).$$

This represents all points on a line through origin.

Example 4.3. Determine whether $(3, 5)$ lies in span of $(1, 2)$ and $(2, 1)$.

Solution:

$$(3, 5) = a(1, 2) + b(2, 1).$$

$$(a + 2b, 2a + b) = (3, 5).$$

Solving gives

$$a = 1, \quad b = 1.$$

Hence belongs to span.

Example 4.4. Find the span of $\{1 + x, x + x^2, 1 + x^2\}$.

Solution:

All polynomials of degree ≤ 2 are generated.

Hence span equals P_2 .

Example 4.5. Determine whether x^2 lies in span of $\{1 + x, 1 + x^2\}$.

Solution:

Assume

$$x^2 = a(1 + x) + b(1 + x^2).$$

Comparing coefficients yields solution.

Hence belongs to span.

4.11 Span of Infinite Sets

For infinite sets S , span is defined as all finite linear combinations.

Example: Polynomial space P is spanned by

$$\{1, x, x^2, x^3, \dots\}.$$

Such sets are called countable generating sets.

4.12 Applications in Engineering

Signal Approximation: Signals are approximated by basis expansions.

Structural Analysis: Force vectors span configuration spaces.

Control Systems: State spaces are spanned by system modes.

Machine Learning: Feature spaces are spanned by basis functions.

4.13 Common Mistakes

- Confusing span with union,
- Ignoring linear combinations,
- Forgetting subspace property,
- Not checking consistency.

4.14 Summary

In this chapter, we studied:

- Definition of span,
- Subspace property,
- Generating sets,
- Matrix interpretation,
- Engineering relevance.

Span provides the foundation for basis and dimension theory.

4.15 Practice Problems

Exercise 4.1. Find the span of $(1, 1)$ and $(2, 3)$.

Exercise 4.2. Determine whether $(4, 6)$ lies in span of $(1, 1)$ and $(2, 2)$.

Exercise 4.3. Find the span of $\{1, x, 1 + x\}$.

Exercise 4.4. Check whether $(1, 2, 3)$ lies in span of $(1, 0, 1)$ and $(0, 1, 1)$.

Exercise 4.5. Find a minimal spanning set for $\{(1, 2), (2, 4), (3, 6)\}$.

Exercise 4.6. Show that $\{1, x, x^2\}$ spans P_2 .

Chapter 5

Dimension of Vector Spaces

5.1 Introduction

The concept of dimension is one of the most important ideas in linear algebra. It measures the “size” or “degree of freedom” of a vector space.

In engineering and applied sciences, dimension determines:

- Number of independent variables,
- Complexity of models,
- Degrees of freedom in mechanical systems,
- Number of parameters in control systems,
- Size of data representations.

This chapter develops the theory of dimension and explains its fundamental role in vector space analysis.

5.2 Motivation

Consider the spaces:

$$\mathbb{R}, \quad \mathbb{R}^2, \quad \mathbb{R}^3.$$

These spaces differ in the number of independent directions.

- $\mathbb{R}$ has one direction,
- $\mathbb{R}^2$ has two directions,
- $\mathbb{R}^3$ has three directions.

This number of independent directions is called the dimension.
Dimension generalizes this idea to arbitrary vector spaces.

5.3 Basis and Dimension

Definition 5.1. Let V be a vector space.

A set $B = \{v_1, v_2, \dots, v_n\}$ is called a **basis** of V if:

1. B spans V ,
2. B is linearly independent.

Definition 5.2. If V has a finite basis consisting of n vectors, then V is called finite-dimensional and n is called the **dimension** of V .

It is denoted by $\dim(V) = n$.

Thus, dimension is defined in terms of bases.

5.4 Existence of Basis

Theorem 5.1. *Every finite-dimensional vector space has a basis.*

Proof. Let V be finite-dimensional.

Then V has a finite spanning set.

Remove dependent vectors one by one.

The remaining set is independent and still spans V .

Hence it forms a basis. □

This result guarantees that dimension is well-defined.

5.5 Invariance of Dimension

Theorem 5.2 (Invariance Theorem). *All bases of a finite-dimensional vector space V contain the same number of vectors.*

Proof. Let B_1 and B_2 be two bases of V .

Since B_1 is independent and B_2 spans V ,

$$|B_1| \leq |B_2|.$$

Similarly,

$$|B_2| \leq |B_1|.$$

Hence,

$$|B_1| = |B_2|.$$

Therefore, all bases have equal cardinality. □

This theorem justifies the definition of dimension.

5.6 Dimension and Spanning Sets

Theorem 5.3. *Let V be a vector space with $\dim(V) = n$.*

Then:

1. *Any spanning set has at least n vectors.*
2. *Any independent set has at most n vectors.*

Proof. Follows directly from the Invariance Theorem and basis properties. $\square$

This theorem provides practical tools for computing dimension.

5.7 Dimension of Subspaces

Theorem 5.4. *Every subspace of a finite-dimensional vector space is finite-dimensional.*

Proof. Let W be a subspace of V .

Any independent set in W is independent in V .

Hence its size is bounded by $\dim(V)$.

Thus W is finite-dimensional. $\square$

Theorem 5.5. *If W is a proper subspace of V , then*

$$\dim(W) < \dim(V).$$

5.8 Matrix Interpretation of Dimension

Let A be an $m \times n$ matrix.

- Dimension of column space = rank of A ,
- Dimension of null space = nullity of A .

Thus, dimension theory connects vector spaces with matrix theory.

5.8.1 Row-Reduced Form

In reduced row echelon form:

- Number of pivots = rank,
- Number of free variables = nullity.

5.9 Geometric Interpretation

5.9.1 Low-Dimensional Spaces

- Dimension 0: single point,
- Dimension 1: line,
- Dimension 2: plane,
- Dimension 3: space.

5.9.2 Higher Dimensions

Higher-dimensional spaces arise in:

- Optimization,
- Data science,
- Machine learning,
- Quantum mechanics.

Though not directly visualizable, their algebraic structure is similar.

5.10 Worked Examples

Example 5.1. Find the dimension of $\mathbb{R}^5$.

Solution:

Standard basis has 5 vectors.

Hence,

$$\dim(\mathbb{R}^5) = 5.$$

Example 5.2. Find the dimension of

$$W = \text{Span}\{(1, 2), (2, 4), (3, 6)\}.$$

Solution:

All vectors are multiples of $(1, 2)$.

Hence only one independent vector.

$$\dim(W) = 1.$$

Example 5.3. Find the dimension of

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}.$$

Solution:

Equation gives one constraint.

So two free variables.

$$\dim(W) = 2.$$

Example 5.4. Find the dimension of P_3 , the space of polynomials of degree ≤ 3 .

Solution:

Basis:

$$\{1, x, x^2, x^3\}.$$

Hence,

$$\dim(P_3) = 4.$$

Example 5.5. Find the dimension of null space of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

Solution:

Rank = 1.

Number of columns = 3.

$$\text{Nullity} = 3 - 1 = 2.$$

Hence null space dimension is 2.

5.11 Applications in Engineering

Mechanical Systems: Dimension represents degrees of freedom.

Electrical Networks: Dimension determines number of independent currents.

Control Theory: State-space dimension controls system order.

Data Science: Dimension measures feature complexity.

Signal Processing: Dimension equals number of independent signals.

5.12 Common Errors

- Confusing size of set with dimension,

- Forgetting independence condition,
- Ignoring subspace constraints,
- Incorrect rank computation.

Careful use of basis and rank avoids these mistakes.

5.13 Summary

In this chapter, we studied:

- Definition of dimension,
- Invariance theorem,
- Relation with basis,
- Subspace dimension,
- Matrix interpretation,
- Engineering applications.

Dimension provides a quantitative description of vector spaces.

5.14 Practice Problems

Exercise 5.1. Find the dimension of

$$W = \text{Span}\{(1, 1, 1), (2, 2, 2), (3, 3, 3)\}.$$

Exercise 5.2. Find the dimension of

$$\{(x, y, z) \in \mathbb{R}^3 : x - y + z = 0\}.$$

Exercise 5.3. Find $\dim(P_4)$.

Exercise 5.4. Find the dimension of column space of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

Exercise 5.5. Determine the dimension of null space of

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}.$$

Exercise 5.6. Show that every subspace of $\mathbb{R}^n$ has dimension $\leq n$.

Chapter 6

Basis and Coordinate Systems

6.1 Introduction

The concept of a basis provides the most efficient description of a vector space. While spanning sets may contain redundant vectors, a basis is both minimal and complete.

In engineering mathematics, a basis:

- Provides coordinate representation,
- Determines dimension,
- Enables efficient computation,
- Forms the foundation of numerical methods,
- Supports signal decomposition and data representation.

This chapter develops the theory of basis and introduces coordinate systems and orthogonalization methods.

6.2 Definition of Basis

Definition 6.1. Let V be a vector space.

A set of vectors $B = \{v_1, v_2, \dots, v_n\}$ is called a **basis** of V if:

1. B spans V ,
2. B is linearly independent.

Thus, a basis is a minimal generating set.

Definition 6.2. If V has a basis consisting of n vectors, then

$$\dim(V) = n.$$

6.3 Characterization of Basis

Theorem 6.1. Let V be finite-dimensional with $\dim(V) = n$.

A set of n vectors in V is a basis if and only if it is:

1. Linearly independent, or
2. Spanning.

Proof. If a set contains n independent vectors in an n -dimensional space, it must span V .

Similarly, if it spans and contains n vectors, it must be independent.

□

This theorem provides a practical test for basis verification.

6.4 Basis Extraction from a Spanning Set

Given a spanning set:

1. Form matrix with vectors as columns.
2. Row reduce to echelon form.
3. Columns corresponding to pivots form a basis.

Theorem 6.2. *Removing dependent vectors from a spanning set yields a basis.*

6.5 Extension of Independent Sets

Theorem 6.3. *Every linearly independent set in a finite-dimensional vector space can be extended to a basis.*

Proof. If independent set does not span V , add vectors from V one by one until spanning is achieved.

The process terminates since dimension is finite. $\square$

This theorem is fundamental in constructive linear algebra.

6.6 Standard Bases

6.6.1 For $\mathbb{R}^n$

The standard basis is:

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, \dots, 0), \quad \dots$$

$$\dim(\mathbb{R}^n) = n.$$

6.6.2 For Polynomial Space P_n

Basis:

$$\{1, x, x^2, \dots, x^n\}.$$

6.6.3 For Matrix Space $M_{m \times n}$

Basis consists of matrices with one entry equal to 1 and others zero.

6.7 Coordinate Representation

Definition 6.3. Let $B = \{v_1, \dots, v_n\}$ be a basis of V .

For $v \in V$,

$$v = a_1v_1 + \dots + a_nv_n.$$

The vector

$$[v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

is called the coordinate vector of v relative to B .

6.7.1 Uniqueness of Coordinates

Theorem 6.4. *Every vector in V has a unique coordinate representation relative to a basis.*

Proof. Follows from independence of basis vectors. □

6.8 Change of Basis

Let $B = \{v_1, \dots, v_n\}$ and $C = \{w_1, \dots, w_n\}$ be two bases of V .

Define matrix P whose columns are coordinates of w_i relative to B .

Then:

$$[v]_B = P[v]_C.$$

6.8.1 Change of Basis Matrix

Matrix P is invertible.

$$[v]_C = P^{-1}[v]_B.$$

This concept is central in diagonalization and linear transformations.

6.9 Gram–Schmidt Orthogonalization

Orthogonal bases simplify computations.

Theorem 6.5. *Given an independent set $\{v_1, \dots, v_n\}$ in an inner product space, there exists an orthogonal basis spanning the same subspace.*

6.9.1 Procedure

Set

$$u_1 = v_1,$$

$$u_2 = v_2 - \text{proj}_{u_1}(v_2),$$

$$u_3 = v_3 - \text{proj}_{u_1}(v_3) - \text{proj}_{u_2}(v_3),$$

and so on.

Where

$$\text{proj}_u(v) = \frac{\langle v, u \rangle}{\langle u, u \rangle} u.$$

The resulting set $\{u_1, \dots, u_n\}$ is orthogonal.

If normalized, it becomes orthonormal.

6.10 Worked Examples

Example 6.1. Find a basis for

$$W = \text{Span}\{(1, 2), (2, 4), (3, 6)\}.$$

Solution:

Vectors are multiples of $(1, 2)$.

Basis:

$$\{(1, 2)\}.$$

Example 6.2. Find a basis for

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}.$$

Solution:

Let

$$x = s, \quad y = t, \quad z = -s - t.$$

General vector:

$$s(1, 0, -1) + t(0, 1, -1).$$

Basis:

$$\{(1, 0, -1), (0, 1, -1)\}.$$

Example 6.3. Find coordinate vector of $(2, 3)$ relative to basis

$$B = \{(1, 1), (1, 2)\}.$$

Solution:

Solve:

$$(2, 3) = a(1, 1) + b(1, 2).$$

$$a + b = 2, \quad a + 2b = 3.$$

$$b = 1, \quad a = 1.$$

$$[(2, 3)]_B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Example 6.4. Apply Gram–Schmidt to

$$\{(1, 1, 0), (1, 0, 1)\}.$$

Solution:

$$u_1 = (1, 1, 0).$$

$$\text{proj}_{u_1}(v_2) = \frac{1}{2}(1, 1, 0).$$

$$u_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

Orthogonal basis obtained.

6.11 Applications in Engineering

Signal Processing: Fourier basis decomposes signals.

Computer Graphics: Basis vectors determine coordinate frames.

Quantum Mechanics: Basis states describe measurable quantities.

Numerical Methods: Orthogonal bases improve numerical stability.

Machine Learning: Principal component analysis uses orthonormal bases.

6.12 Common Errors

- Assuming spanning implies independence,
- Forgetting uniqueness of coordinates,
- Incorrect pivot identification,
- Errors in projection formula.

6.13 Summary

In this chapter, we developed:

- Definition of basis,
- Characterization theorems,
- Basis extraction,
- Coordinate systems,
- Change of basis,

- Gram–Schmidt process.

Basis theory forms the backbone of vector space analysis.

6.14 Practice Problems

Exercise 6.1. Find a basis for

$$\text{Span}\{(1, 0, 1), (2, 1, 3), (3, 1, 4)\}.$$

Exercise 6.2. Find the coordinate vector of $(3, 4)$ relative to

$$\{(1, 1), (1, 2)\}.$$

Exercise 6.3. Extend $\{(1, 0, 1)\}$ to a basis of $\mathbb{R}^3$.

Exercise 6.4. Apply Gram–Schmidt to

$$\{(1, 1, 1), (1, 0, 1), (0, 1, 1)\}.$$

Exercise 6.5. Find change of basis matrix from standard basis to

$$\{(1, 1), (1, -1)\}.$$

Exercise 6.6. Prove that any basis of $\mathbb{R}^n$ contains exactly n vectors.

Chapter 7

Subspaces, Null Space, Column Space, and Rank

7.1 Introduction

Subspaces and matrix spaces form the structural backbone of linear algebra. They describe solution spaces of systems, feasible regions in optimization, and state spaces in engineering models.

The null space, column space, and row space of a matrix are called the **fundamental subspaces**. Their dimensions are governed by the rank of the matrix.

This chapter develops the theory and applications of these concepts.

7.2 Subspaces

7.2.1 Definition

Definition 7.1. A nonempty subset W of a vector space V is called a **subspace** if:

1. $u + v \in W$ for all $u, v \in W$,
2. $cu \in W$ for all $u \in W$ and $c \in F$,
3. $0 \in W$.

7.2.2 Subspace Test

Theorem 7.1 (Subspace Test). *A nonempty subset W of V is a subspace if and only if:*

$$u, v \in W, c \in F \Rightarrow u + cv \in W.$$

Proof. Assume W is a subspace. Then closure under addition and scalar multiplication gives result. Conversely, setting $c = 0$ and $u = v$ yields required properties. $\square$

7.2.3 Examples of Subspaces

- Solution spaces of homogeneous systems,
- Span of vectors,
- Polynomial spaces,
- Null spaces of matrices.

7.3 Null Space

7.3.1 Definition

Definition 7.2. Let A be an $m \times n$ matrix.

The **null space** of A is defined as

$$N(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

Null space represents the solution space of the homogeneous system.

7.3.2 Properties

Theorem 7.2. $N(A)$ is a subspace of $\mathbb{R}^n$.

Proof. If $Ax = 0$ and $Ay = 0$, then

$$A(x + y) = 0, \quad A(cx) = 0.$$

Hence closure holds. □

7.4 Column Space

7.4.1 Definition

Definition 7.3. Let $A = [a_1 \ a_2 \ \cdots \ a_n]$.

The **column space** of A is

$$\text{Col}(A) = \text{Span}\{a_1, \dots, a_n\}.$$

Column space consists of all vectors obtainable as Ax .

7.4.2 Interpretation

$Ax = b$ is solvable if and only if $b \in \text{Col}(A)$.

Thus, column space characterizes consistency.

7.5 Row Space

7.5.1 Definition

Definition 7.4. The **row space** of A is the span of its rows.

Row space equals column space of A^T .

7.6 Fundamental Subspaces

For $A \in \mathbb{R}^{m \times n}$:

- Null space: $N(A) \subseteq \mathbb{R}^n$,
- Column space: $\text{Col}(A) \subseteq \mathbb{R}^m$,
- Row space: $\text{Row}(A) \subseteq \mathbb{R}^n$,
- Left null space: $N(A^T) \subseteq \mathbb{R}^m$.

These four spaces describe matrix behavior.

7.7 Rank of a Matrix

7.7.1 Definition

Definition 7.5. The **rank** of a matrix A is the dimension of its column space.

Equivalently, it is dimension of row space.

7.7.2 Rank via Row Reduction

Theorem 7.3. *The rank of a matrix equals the number of pivots in its reduced row echelon form.*

Thus rank is computed by Gaussian elimination.

7.8 Properties of Rank

Theorem 7.4. *For any matrix A :*

1. $\text{rank}(A) = \text{rank}(A^T)$,
2. $\text{rank}(A) \leq \min(m, n)$,
3. $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.

7.9 Relation Between Fundamental Subspaces

Theorem 7.5. *The row space and null space are orthogonal complements in $\mathbb{R}^n$.*

Theorem 7.6. *The column space and left null space are orthogonal complements in $\mathbb{R}^m$.*

These relations are essential in numerical analysis.

7.10 Worked Examples

Example 7.1. Find null space of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

Solution:

Row reduce:

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

Equation:

$$x_1 + 2x_2 + 3x_3 = 0.$$

Free variables: $x_2 = s$, $x_3 = t$.

$$x = (-2s - 3t, s, t).$$

Basis obtained.

Example 7.2. Find a basis for column space of

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 1 \end{bmatrix}.$$

Solution:

Row reduce and identify pivot columns.

First and third columns form basis.

Example 7.3. Find rank of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

Solution:

Row reduction gives two pivots.

Rank = 2.

Example 7.4. Determine whether $Ax = b$ is consistent for

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

Solution:

b lies in column space.

System is consistent.

Example 7.5. Find basis for row space of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

Solution:

RREF has one nonzero row.

Basis = that row.

7.11 Applications in Engineering

Circuit Analysis: Null space represents loop currents.

Structural Mechanics: Column space models displacement modes.

Control Theory: Rank determines controllability.

Signal Processing: Null space filters noise.

Optimization: Constraint spaces are subspaces.

7.12 Common Errors

- Confusing null space with column space,
- Taking wrong pivot columns,
- Ignoring free variables,
- Miscounting rank.

7.13 Summary

This chapter developed:

- Subspace theory,
- Null, column, and row spaces,
- Rank computation,
- Fundamental subspaces,
- Engineering relevance.

These ideas prepare for Rank–Nullity Theorem.

7.14 Practice Problems

Exercise 7.1. Find null space of

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}.$$

Exercise 7.2. Find basis for column space of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

Exercise 7.3. Find rank of

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

Exercise 7.4. Determine whether

$$Ax = b$$

is consistent.

Exercise 7.5. Prove that $N(A)$ is always a subspace.

Exercise 7.6. Show that $\text{rank}(A) = \text{rank}(A^T)$.

Chapter 8

The Rank–Nullity Theorem

8.1 Introduction

One of the most important results in linear algebra is the **Rank–Nullity Theorem**. It establishes a fundamental relationship between the solution space of a linear system and the information contained in its coefficient matrix.

This theorem explains how the total number of variables in a system is divided into:

- Independent (leading) variables,
- Dependent (free) variables.

In engineering and applied sciences, this theorem plays a central role in:

- System analysis,
- Network theory,

- Control systems,
- Optimization,
- Data modeling.

This chapter develops the theory, proof, and applications of the Rank–Nullity Theorem.

8.2 Preliminaries

Let A be an $m \times n$ matrix.

Recall the following definitions:

- $\text{Rank}(A)$: Dimension of column space,
- $\text{Nullity}(A)$: Dimension of null space,
- $N(A) = \{x : Ax = 0\}$.

The null space represents all homogeneous solutions, while the column space represents attainable outputs.

8.3 Statement of the Rank–Nullity Theorem

Theorem 8.1 (Rank–Nullity Theorem). *Let A be an $m \times n$ matrix over a field F .*

Then,

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Equivalently, for a linear transformation $T : V \rightarrow W$,

$$\dim(\ker T) + \dim(\text{Im } T) = \dim(V).$$

This theorem applies to all finite-dimensional vector spaces.

8.4 Interpretation of the Theorem

Let $Ax = 0$ be a homogeneous system.

- Rank = number of leading variables,
- Nullity = number of free variables,
- n = total variables.

Thus,

$$\text{Total Variables} = \text{Independent Variables} + \text{Free Variables}.$$

This explains why every system decomposes into constrained and unconstrained components.

8.5 Proof Using Row Reduction

Proof. Let A be reduced to its row echelon form R by elementary row operations.

Row operations do not change rank or nullity.

Suppose R has r pivots.

Then:

- Number of leading variables = r ,
- Number of free variables = $n - r$.

Hence,

$$\text{rank}(A) = r, \quad \text{nullity}(A) = n - r.$$

Therefore,

$$\text{rank}(A) + \text{nullity}(A) = n.$$

□

This proof emphasizes computational aspects.

8.6 Proof Using Linear Transformations

Let $T : V \rightarrow W$ be a linear transformation.

Define:

$$\ker(T) = \{v : T(v) = 0\}, \quad \operatorname{Im}(T) = \{T(v) : v \in V\}.$$

Proof. Let

$$\{k_1, \dots, k_m\}$$

be a basis of $\ker(T)$.

Extend this to a basis of V :

$$\{k_1, \dots, k_m, v_{m+1}, \dots, v_n\}.$$

Then,

$$\{T(v_{m+1}), \dots, T(v_n)\}$$

is a basis of $\operatorname{Im}(T)$.

Hence,

$$\dim(\ker T) = m, \quad \dim(\operatorname{Im} T) = n - m.$$

So,

$$\dim(\ker T) + \dim(\operatorname{Im} T) = n.$$

□

This proof emphasizes structural aspects.

8.7 Geometric Interpretation

8.7.1 In $\mathbb{R}^2$

If $A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$:

- Rank = 2 $\rightarrow$ full plane,
- Rank = 1 $\rightarrow$ line,
- Rank = 0 $\rightarrow$ point.

Nullity adjusts accordingly.

8.7.2 In $\mathbb{R}^3$

Possible cases:

Rank	Nullity	Mapping
3	0	Invertible
2	1	Plane collapse
1	2	Line collapse
0	3	Zero map

8.8 Relation to Consistency of Systems

Consider $Ax = b$.

Theorem 8.2. *The system $Ax = b$ is consistent if and only if*

$$b \in \text{Col}(A).$$

If consistent, solution set is:

$$x = x_p + x_h,$$

where:

- x_p is particular solution,
- $x_h \in N(A)$.

Dimension of solution space equals nullity.

8.9 Worked Examples

Example 8.1. Let A be 4×6 with rank 3. Find nullity.

Solution:

$$3 + \text{nullity} = 6$$

$$\text{nullity} = 3.$$

Example 8.2. Find rank and nullity of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

Solution:

Row reduce:

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}.$$

Rank = 2.

$$2 + \text{nullity} = 3 \Rightarrow \text{nullity} = 1.$$

Example 8.3. Find dimension of solution space of

$$Ax = 0$$

where A is 5×8 with rank 4.

Solution:

$$4 + \text{nullity} = 8$$

$$\text{nullity} = 4.$$

Example 8.4. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with

$$T(x, y, z) = (x + y, y + z).$$

Find rank and nullity.

Solution:

Matrix:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Rank = 2.

$$2 + \text{nullity} = 3 \Rightarrow \text{nullity} = 1.$$

Example 8.5. Find dimension of kernel of

$$T : P_3 \rightarrow P_2, \quad T(p) = p'.$$

Solution:

Kernel = constant polynomials.

Dimension = 1.

P_3 has dimension 4.

Rank = 3.

8.10 Applications in Engineering

Control Systems: Nullity measures uncontrollable modes.

Network Theory: Rank determines solvability of circuits.

Signal Processing: Null space filters noise components.

Optimization: Rank controls constraint redundancy.

Machine Learning: Rank measures feature information.

8.11 Common Errors

- Forgetting column count,
- Confusing rows with columns,
- Miscounting pivots,
- Applying theorem to non-finite spaces.

8.12 Summary

In this chapter, we developed:

- Statement of Rank–Nullity Theorem,
- Algebraic and geometric proofs,
- Relation to systems,
- Engineering relevance.

The Rank–Nullity Theorem unifies all major concepts of linear algebra.

8.13 Practice Problems

Exercise 8.1. Let A be 3×7 with rank 2. Find nullity.

Exercise 8.2. Find rank and nullity of

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \end{bmatrix}.$$

Exercise 8.3. Find dimension of kernel of

$$T : \mathbb{R}^4 \rightarrow \mathbb{R}^3.$$

Exercise 8.4. Prove Rank–Nullity Theorem using basis extension.

Exercise 8.5. Show that invertible matrices have nullity zero.

Exercise 8.6. Find dimension of solution space of

$$Ax = b$$

if $\text{rank}(A) = 3$ and $n = 6$.

Formula Sheet: Linear Algebra

Chapter 1: Linear Combinations

Linear Combination

$$v = a_1v_1 + a_2v_2 + \cdots + a_nv_n$$

Matrix Form

$$Ax = v \quad \text{where } A = [v_1 \ v_2 \ \cdots \ v_n]$$

Existence

$$v \in \text{Span}\{v_1, \dots, v_n\} \iff Ax = v \text{ is consistent}$$

Uniqueness

$$\text{Unique representation} \iff \{v_i\} \text{ is independent.}$$

Chapter 2: Linear Independence

Definition

$$a_1v_1 + \cdots + a_nv_n = 0 \Rightarrow a_1 = \cdots = a_n = 0$$

Matrix Test

$Ax = 0$ has only trivial solution $\iff$ independent

Determinant Test (Square Matrix)

$$\det(A) \neq 0 \iff \text{independent}$$

Rank Criterion

$$\text{rank}(A) = n \iff \text{independent}$$

Basic Results

- Set containing 0 is dependent
- More than n vectors in $\mathbb{R}^n$ are dependent
- Subset of independent set is independent

Chapter 3: Span

Definition

$$\text{Span}\{v_1, \dots, v_n\} = \left\{ \sum a_i v_i \right\}$$

Subspace Property

$\text{Span}(S)$ is always a subspace

Generating Set

$$\text{Span}(S) = V \Rightarrow S \text{ generates } V$$

Column Space

$$\text{Span}\{v_1, \dots, v_n\} = \text{Col}(A)$$

Chapter 4: Dimension

Definition

$$\dim(V) = \text{number of vectors in a basis}$$

Invariance Theorem

All bases have same number of vectors.

Dimension Theorem

$$\text{Independent set} \leq \dim(V)$$

$$\text{Spanning set} \geq \dim(V)$$

Subspace Dimension

$$W \subseteq V \Rightarrow \dim(W) \leq \dim(V)$$

Common Spaces

$$\dim(\mathbb{R}^n) = n$$

$$\dim(P_n) = n + 1$$

$$\dim(M_{m \times n}) = mn$$

Chapter 5: Basis and Coordinate Systems

Basis

A set B is a basis if:

$$\text{Span}(B) = V \quad \text{and} \quad B \text{ is independent}$$

Basis Test

n vectors in n -dim space:

$$\text{Independent} \iff \text{Basis} \iff \text{Spanning}$$

Coordinate Representation

$$v = a_1v_1 + \cdots + a_nv_n$$

$$[v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

Change of Basis

$$[v]_B = P[v]_C$$

$$[v]_C = P^{-1}[v]_B$$

Gram–Schmidt Process

$$u_1 = v_1$$

$$u_2 = v_2 - \text{proj}_{u_1}(v_2)$$

$$u_3 = v_3 - \text{proj}_{u_1}(v_3) - \text{proj}_{u_2}(v_3)$$

Projection Formula

$$\text{proj}_u(v) = \frac{\langle v, u \rangle}{\langle u, u \rangle} u$$

Normalization

$$e_i = \frac{u_i}{\|u_i\|}$$

Chapter 6: Subspaces, Null Space, Column Space, Rank

Subspace Test

$$u, v \in W, c \in F \Rightarrow u + cv \in W$$

Null Space

$$N(A) = \{x : Ax = 0\}$$

Column Space

$$\text{Col}(A) = \text{Span of columns of } A$$

Row Space

$$\text{Row}(A) = \text{Span of rows of } A$$

Rank

$$\text{rank}(A) = \dim(\text{Col}(A))$$

Rank via RREF

rank = number of pivots

Fundamental Subspaces

- $N(A)$: Null space
- $\text{Col}(A)$: Column space
- $\text{Row}(A)$: Row space
- $N(A^T)$: Left null space

Orthogonality

$$\text{Row}(A) \perp N(A)$$

$$\text{Col}(A) \perp N(A^T)$$

Chapter 7: Rank–Nullity Theorem

Matrix Form

$$\text{rank}(A) + \text{nullity}(A) = n$$

(A is $m \times n$)

Transformation Form

For $T : V \rightarrow W$,

$$\dim(\ker T) + \dim(\text{Im } T) = \dim(V)$$

Nullity

$$\text{nullity}(A) = \dim(N(A))$$

Solution Space of

$$Ax = b$$

If consistent:

$$x = x_p + x_h$$

where

$$Ax_p = b, \quad x_h \in N(A)$$

Dimension of Solution Space

$$\dim(\text{solution space}) = \text{nullity}(A)$$

Invertibility

$$A \text{ invertible} \iff \text{rank}(A) = n \iff \text{nullity}(A) = 0$$

Important Identities

Rank Properties

$$\text{rank}(A) = \text{rank}(A^T)$$

$$\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$$

System Consistency

$$Ax = b \text{ consistent} \iff b \in \text{Col}(A)$$

Dimension of Polynomial Space

$$\dim(P_n) = n + 1$$

Standard Bases

$$\mathbb{R}^n : \{e_1, \dots, e_n\}$$

$$P_n : \{1, x, \dots, x^n\}$$

$$M_{m \times n} : E_{ij}$$

Exam Quick Reference

- Independent $\Rightarrow$ unique representation
- Basis = Independent + Spanning
- Rank = Number of pivots
- Nullity = Free variables
- Span = Column space
- Dimension = Size of basis
- Rank–Nullity = Column count

References

1. D. Poole, Introduction to Linear Algebra.
2. Gilbert Strang, Introduction to Linear Algebra, 5th Edition.